THE TRIANGLE OEIS A211343: ITS POSITIVE ENTRIES ARE THE SMALLEST PARTS OF PARTITIONS INTO CONSECUTIVE PARTS, AND THE LAST ONE IN ROW n LIES IN COLUMN A109814(n)

MATH0

Abstract. Let $T(n,k)$ be the triangle OEIS A211343, in which column k lists the positive integers interleaved with $k-1$ zeros, starting in row $k(k+1)/2$. We prove the two conjectures recorded in the entry by Omar E. Pol in April 2017: if $T(n,k)>0$, then $T(n,k)$ is the smallest part of the partition of n into k consecutive parts; and the last positive entry of row n lies in column A109814(n), the largest k such that n is a sum of k consecutive positive integers. Both follow from the identity $m+(m+1)+\cdots+(m+k-1)=km+k(k-1)/2$, and the key facts are already recorded, partly with proofs, in neighbouring OEIS entries (D. A. Corneth, P. Munn, M. Alekseyev); we therefore classify the result as the first formal proof of a known result, not as a new result. The proof is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the 94 listed terms of A211343 and the first 40 terms of A109814.

1. Introduction

For $k\ge 1$ write $\Delta_k=k(k+1)/2$ for the k-th triangular number. The OEIS entry A211343 [10], contributed by Omar E. Pol on February 5, 2013, is the irregular triangle $T(n,k)$ ($n\ge 1$, $1\le k$, $\Delta_k\le n$) in which column k begins in row Δ_k and reads, going down,

$$1,\ \underbrace{0,\dots,0}_{k-1},\ 2,\ \underbrace{0,\dots,0}_{k-1},\ 3,\ \dots$$

Row n thus has length A003056(n), the largest k with $\Delta_k\le n$. Rows 15 to 18 are $[15,7,4,0,1]$, $[16,0,0,0,0]$, $[17,8,0,0,0]$, $[18,0,5,3,0]$. On April 30, 2017 Pol added two conjectures to the entry (quoted verbatim in Section 2): a positive entry $T(n,k)$ is the smallest part of the partition of n into k consecutive parts, and the last positive entry of row n is in column A109814(n) [7], the largest k for which n is a sum of k consecutive positive integers. For example, the partitions of 15 into consecutive parts are 15, $8+7$, $6+5+4$ and $5+4+3+2+1$, with smallest parts $15,7,4,1$ in columns $1,2,3,5$, and A109814$(15)=5$. Both lines are still labelled "Conjecture" in the revision of the entry we used (#54, March 12, 2021). We prove both conjectures (Theorem 1). The key fact (Proposition 5) is that for $n\ge 1$ the entry $T(n,k)$ is positive exactly when $n=m+(m+1)+\cdots+(m+k-1)$ for some $m\ge 1$, and then $T(n,k)=m$.

Prior work and status. The substance of the result is already in the literature and in the OEIS.

- Writing $n=m+(m+1)+\cdots+(m+k-1)$ as $2n=k(2m+k-1)$, in which exactly one factor is odd, shows that the partitions of n into consecutive parts correspond to the odd divisors of n. This classical enumeration is the one-run case of Sylvester's refinement of Euler's partition theorem [5]; it is called Sylvester's enumeration in A001227 [6]; see also Apostol [1]. In A109814 [7], J. Spies (August 16, 2005, after D. W. Wilson) uses $2n=k(2m+k-1)$ to compute A109814.

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- In A285898 [13], D. A. Corneth noted on April 28, 2017 that n has a partition into exactly k consecutive parts if and only if $(n - \binom{k}{2})/k$ is a positive integer, which is the starting part m. In the same entry S. Cadot remarked (January 22, 2023) that there is at most one such partition.
- For the companion triangle A196020 [9] (odd numbers in place of the positive integers), M. Alekseyev gave on November 18, 2013 the closed form $2n/k - k$ on its support. Combined with the relation $T(n,k) = \lfloor (1 + \text{A196020}(n,k))/2 \rfloor$ in A211343, this yields $T(n,k) = n/k - (k-1)/2$ on the support, which is the smallest part m above. H. F. W. Hoft (A237048 [11], October 23, 2014) described the support of these triangles via the odd divisors of n.
- For the companion triangle A286013 [14] (column k lists $k, k+1, \ldots$ interleaved with zeros), whose positive entries are $T(n,k)+k-1$, P. Munn gave on June 18, 2017 a proof, by trapezoidal dot patterns, of the analogous conjecture for largest parts: the positive entries of column k correspond one-to-one to the partitions of n into k consecutive parts. The same argument proves Conjecture 1. In A261699 [12] (July 30, 2017) Munn states explicitly that $\text{A211343}(n,k) = |n/d - d/2| + 1/2$ is the smallest part of the partition of n attached to the odd divisor d (with some steps left as "it can be shown").
- Other entries by Pol state both conjectures as facts. A379632 [15] (December 31, 2024), the smallest parts of the partitions of n into consecutive parts, is cross-referenced as "Positive terms of A211343" with "Right border gives A118235", where A118235 [8] (R. Zumkeller, 2006) is the smallest part of the longest such partition. The cross-references of A286013, which has the same support as A211343, include "The last positive term in row n is in column A109814(n)".

The math0 referees found no paper addressing A211343 by name. Since the conjectures follow from these identities in a few lines, we classify our result as the *first formal proof of a known result.*

2. Statement

The OEIS conjectures. The name of A211343 [10] (Omar E. Pol, February 5, 2013; offset 1) is, verbatim,

> Triangle read by rows: T(n,k), n >= 1, k >= 1, in which column k lists the positive integers interleaved with k-1 zeros, and the first element of column k is in row k(k+1)/2.

Its COMMENTS section contains, verbatim, under the heading "From *Omar E. Pol*, Apr 30 2017":

> Conjecture 1: T(n,k) is the smallest part of the partition of n into k consecutive parts, if T(n,k) > 0.
> Conjecture 2: the last positive integer in the row n is in the column A109814(n).

The sequence A109814 [7] (David W. Wilson) is "a(n) is the largest k such that n can be written as sum of k consecutive positive integers."

The theorem. For $k \geq 1$ and $j \geq 0$ let $c_k(j) = j/k + 1$ if $k \mid j$ and $c_k(j) = 0$ otherwise; this is column k read from its first row. Row n consists of the columns k with $1 \leq k$ and $\Delta_k \leq n$, and for these $T(n,k) = c_k(n - \Delta_k)$. A *partition of n into k consecutive parts* is a multiset of positive integers with sum n of the form $\{m, m+1, \ldots, m+k-1\}$.

Theorem 1. *Let $n \geq 1$.*

(1) *If $1 \leq k$, $\Delta_k \leq n$ and $T(n,k) > 0$, then n has a partition into k consecutive parts, and $T(n,k)$ is the smallest part of every such partition.*

(2) *Let $A(n)$ be the largest k such that n is a sum of k consecutive positive integers. Then $A(n)$ is the largest k with $1 \le k$, $\Delta_k \le n$ and $T(n,k) > 0$.*

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A211343.lean`, reviewed for fidelity, and frozen (SHA-256 `bc8da2da...897e`) before any proof attempt. The definitions and the statement read as follows (verbatim; the copyright header, the import, the module docstring and all docstrings are omitted):

```
namespace Math0.Statements.A211343
open Finset
def column (k j : ℕ) : ℕ := if j % k = 0 then j / k + 1 else 0
def InRow (n k : ℕ) : Prop := 1 ≤ k ∧ k * (k + 1) / 2 ≤ n
instance (n k : ℕ) : Decidable (InRow n k) := by unfold InRow; infer_instance
def T (n k : ℕ) : ℕ := if InRow n k then column k (n - k * (k + 1) / 2) else 0
def row (n : ℕ) : List ℕ := ((List.range (n + 1)).filter (fun k => decide (InRow n k))).map (T
    n)
def a (i : ℕ) : ℕ := ((List.range i).flatMap (fun j => row (j + 1))).getD (i - 1) 0
def HasConsecutiveParts {n : ℕ} (k : ℕ) (p : n.Partition) : Prop :=
  ∃ m, p.parts = (Multiset.range k).map (fun i => m + i)
def IsConsecSum (n k : ℕ) : Prop := ∃ m, 0 < m ∧ ∑ i ∈ range k, (m + i) = n
def IsConsecSumBdd (n k : ℕ) : Prop := ∃ m ∈ Finset.Icc 1 n, ∑ i ∈ Finset.range k, (m + i) = n
instance (n : ℕ) : DecidablePred (IsConsecSumBdd n) := fun k => by
  unfold IsConsecSumBdd; infer_instance
def A109814 (n : ℕ) : ℕ := Nat.findGreatest (IsConsecSumBdd n) n
def statement : Prop :=
  (∀ n k : ℕ, 1 ≤ n → InRow n k → 0 < T n k →
    (∃ p : n.Partition, HasConsecutiveParts k p) ∧
    ∀ p : n.Partition, HasConsecutiveParts k p → IsLeast {x | x ∈ p.parts} (T n k)) ∧
  (∀ n : ℕ, 1 ≤ n → IsGreatest {k | InRow n k ∧ 0 < T n k} (A109814 n))
end Math0.Statements.A211343
```

The file also proves the following (statements shown, proofs omitted; the lists of 94 terms, 24 rows and 40 terms on the right-hand sides of the data checks are elided):

```
theorem le_of_inRow {n k : ℕ} (h : InRow n k) : k ≤ n
theorem a_data : (List.range 94).map (fun i => a (i + 1)) =
    [1, 2, 3, 1, 4, 0, 5, 2, 6, 0, 1, 7, 3, 0, 8, 0, 0, 9, 4, 2, ..., 24, 0, 7, 0, 0, 0]
theorem row_data : (List.range 24).map (fun n => row (n + 1)) = [[1], [2], [3, 1], ...]
theorem A109814_isGreatest {n : ℕ} (hn : 1 ≤ n) :
    IsGreatest {k | IsConsecSum n k} (A109814 n)
theorem A109814_data : (List.range 40).map (fun i => A109814 (i + 1)) =
    [1, 1, 2, 1, 2, 3, 2, 1, 3, 4, 2, 3, 2, 4, 5, 1, 2, 4, 2, 5, ..., 8, 2, 4, 6, 5]
```

How the definitions encode the OEIS entry. *The triangle* follows the name of the entry, not any conjectured formula. `column k j` is $c_k(j)$: for $k \ge 1$, `j % k = 0` means $k \mid j$. `InRow n k` says that (n,k) is a position of the triangle; `k * (k + 1) / 2` is exactly Δ_k, and under `InRow` the truncated subtraction `n - k * (k + 1) / 2` is the true difference. Outside the triangle `T n k` is 0, a junk value that `statement` never uses, since both conjuncts assume `InRow n k`. `row n` lists $T(n,1),\dots,T(n,\mathrm{A003056}(n))$ (by `le_of_inRow`, no column exceeds n), and `a` is the triangle read by rows with offset 1.

Data checks. `a_data` states that $a(1),\dots,a(94)$ are the 94 terms in the DATA section of A211343 (rows 1 to 24), `row_data` that rows 1 to 24 agree with the first 24 rows displayed in the EXAMPLE section (which shows rows 1 to 28), and `A109814_data` that `A109814` reproduces the first 40 listed terms of A109814. All three are proved by `decide +kernel`: the kernel

evaluates both sides and checks that they are equal. This anchors the definitions to the entry independently of the conjectures.

Partitions. A partition of n is Mathlib's `Nat.Partition n`, a multiset `parts` of positive integers with sum n. `HasConsecutiveParts k p` says that `p.parts` is $\{m+i : 0 \le i < k\}$ for some m. "$T(n,k)$ is the smallest part of the partition" is read as: such a partition exists, and $T(n,k)$ is the least part of every such partition (`IsLleast`). Only the stated direction is claimed; the partition is in fact unique (Remark 7). The hypotheses $1 \le n$ and `InRow n k` follow from $T(n,k) > 0$.

A109814. `IsConsecSum n k` says that $n = \sum_{i<k}(m+i)$ for some $m \ge 1$, with no bounds; this is the OEIS definition. For decidability, `A109814 n` is computed with `Nat.findGreatest`, which returns the largest $k \le n$ satisfying the bounded predicate `IsConsecSumBdd n` (start $m \in [1,n]$), or 0 if there is none; by Lemma 6 (`A109814_isGreatest`, proved in the statement file) the bounds change nothing for $n \ge 1$. *Conjecture 2* is encoded as: `A109814 n` is the greatest column k of row n with $T(n,k) > 0$; this set contains $k = 1$, as $T(n,1) = n$.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A211343.lean`; the Lean names of the lemmas are given in brackets. Throughout, $S_k = 0 + 1 + \cdots + (k-1) = k(k-1)/2$.

Lemma 2 (`tri_eq`, `consec_sum`, `multiset_consec_sum`)**.** *For all $k, m \ge 0$: $\Delta_k = S_k + k$, and both $\sum_{i<k}(m+i)$ and the sum of the multiset $\{m+i : 0 \le i < k\}$ equal $km + S_k$.*

Proof. By Gauss' formula (`Finset.sum_range_id`), $S_{k+1} = (k+1)k/2 = \Delta_k$, and $S_{k+1} = S_k + k$. The second identity is linearity of the sum. The multiset sum is the same sum, since `Finset.range k` has underlying multiset `Multiset.range k`. □

Lemma 3 (`eq_of_T_pos`)**.** *If $1 \le k$, $\Delta_k \le n$ and $T(n,k) > 0$, then $n = k\,T(n,k) + S_k$.*

Proof. Let $j = n - \Delta_k \ge 0$. By definition $T(n,k) = c_k(j)$, and $c_k(j) > 0$ rules out the branch $c_k(j) = 0$; so $k \mid j$ and $T(n,k) = j/k + 1$. Hence, using Lemma 2,

$$k\,T(n,k) + S_k = k \cdot \frac{j}{k} + k + S_k = j + \Delta_k = n.$$ □

Lemma 4 (`T_of_eq`)**.** *If $k \ge 1$, $m \ge 1$ and $n = km + S_k$, then $1 \le k$, $\Delta_k \le n$ and $T(n,k) = m$.*

Proof. Since $km \ge k$, we have $\Delta_k = S_k + k \le S_k + km = n$. Moreover $j = n - \Delta_k = km - k = k(m-1)$, so $k \mid j$ and $T(n,k) = c_k(j) = (m-1) + 1 = m$. □

Proposition 5 (`inRow_T_pos_iff`)**.** *Let $n \ge 1$ and $k \ge 0$. Then*

$$\big(1 \le k,\ \Delta_k \le n \text{ and } T(n,k) > 0\big) \iff \exists\, m \ge 1 :\ \sum_{i<k}(m+i) = n,$$

and in this case $m = T(n,k)$.

Proof. If the left side holds, take $m = T(n,k) \ge 1$; then $\sum_{i<k}(m+i) = km + S_k = n$ by Lemmas 2 and 3. Conversely, let $m \ge 1$ with $\sum_{i<k}(m+i) = n$, i.e. $km + S_k = n$. If $k = 0$ the sum is empty and $n = 0$, contradicting $n \ge 1$; so $k \ge 1$, and Lemma 4 gives the left side and $T(n,k) = m$. Given $k \ge 1$, the value of m is determined by $km + S_k = n$. □

Lemma 6 (`A109814_isGreatest`, in the statement file)**.** *For $n \ge 1$, `A109814 n` is the greatest k such that $n = \sum_{i<k}(m+i)$ for some $m \ge 1$. In particular it equals $A(n)$ of Theorem 1.*

Proof. Write $P(k)$ for the bounded predicate: $\sum_{i<k}(m+i) = n$ for some $m \in [1, n]$. Since $P(1)$ holds with $m = n$ and $1 \le n$, `Nat.findGreatest P n` is the largest $k \le n$ with $P(k)$, and it satisfies P, hence the unbounded property. Conversely, if $\sum_{i<k}(m+i) = n$ with $m \ge 1$, then $k \le n$ because each of the k summands is at least 1; $k \ge 1$ because $n \ge 1$; and $m \le n$ because m is the summand $i = 0$. So $P(k)$ holds with $k \le n$, and $k \le$ `A109814 n` (`Nat.le_findGreatest`). □

Proof of Theorem 1. (1) Let $m = T(n, k) \ge 1$. By Lemma 3, $n = km + S_k$.

Existence. The multiset $P = \{m + i : 0 \le i < k\}$ has positive elements, since $m \ge 1$, and its sum is $km + S_k = n$ by Lemma 2. So P is a partition of n, and it has k consecutive parts starting at m.

Smallest part. Let p be any partition of n whose parts are $\{m' + i : 0 \le i < k\}$. Summing its parts and using Lemma 2, $km' + S_k = n = km + S_k$, so $km' = km$ and $m' = m$ because $k \ge 1$. Since $0 < k$, the part $m + 0 = m$ occurs in p, and every part $m + i$ of p is at least m. Hence m is the least part of p.

(2) By Proposition 5, for $n \ge 1$ the set of columns k of row n with $T(n, k) > 0$ equals the set of k such that n is a sum of k consecutive positive integers. By Lemma 6, A109814(n) is the greatest element of the latter set, hence of the former. □

Remark 7. Proposition 5, whose ingredients are formalized (`inRow_T_pos_iff` and `T_of_eq`), also gives the converse of Conjecture 1: every partition of n into k consecutive parts, $k \ge 1$, starts at $m = T(n, k) > 0$. This converse is not part of `statement` and is not stated as a separate Lean theorem. Thus the positive entries of row n are exactly the smallest parts of the partitions of n into consecutive parts, one for each number of parts, and each such partition is unique. On the support, $T(n, k) = n/k - (k-1)/2$, in agreement with Alekseyev's formula (Section 1). That the number of positive entries of row n is the number of odd divisors of n, as stated in A211343 [10], then follows from the classical enumeration recalled in Section 1; this count is not formalized here.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A211343.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A211343.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A211343.result`, of type `Math0.Statements.A211343.statement`. The six auxiliary lemmas are private. Their statements read verbatim as follows (all proofs, the header, the imports and the docstrings omitted; the 33-line proof of `result` carries out the proof of Theorem 1 in the order given in Section 3):

```
namespace Math0.Frontier.A211343
open Math0.Statements.A211343
open Finset
private lemma tri_eq (k : ℕ) : k * (k + 1) / 2 = (∑ i ∈ range k, i) + k
private lemma consec_sum (m k : ℕ) :
    ∑ i ∈ range k, (m + i) = k * m + ∑ i ∈ range k, i
private lemma multiset_consec_sum (m k : ℕ) :
    ((Multiset.range k).map (fun i => m + i)).sum = k * m + ∑ i ∈ range k, i
private lemma eq_of_T_pos {n k : ℕ} (h : InRow n k) (hT : 0 < T n k) :
    n = k * T n k + ∑ i ∈ range k, i
private lemma T_of_eq {n k m : ℕ} (hk : 1 ≤ k) (hm : 1 ≤ m)
    (hn : n = k * m + ∑ i ∈ range k, i) : InRow n k ∧ T n k = m
private lemma inRow_T_pos_iff {n k : ℕ} (hn : 1 ≤ n) :
    (InRow n k ∧ 0 < T n k) ↔ IsConsecSum n k
theorem result : statement
end Math0.Frontier.A211343
```

```
#math0_settles Math0.Frontier.A211343.result Math0.Statements.A211343.statement
```

The arithmetic is closed by `omega`, with the sums S_k and products such as $kT(n,k)$ as atoms. No computation is involved in the proof; the only kernel evaluations are the data checks of the statement file. Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`.

Audit. The command `#math0_settles` (`lean/Math0/Audit.lean`), used for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata annotations, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A211343.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A211343.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A211343.result' proves 'Math0.Statements.A211343.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use, since the statement is a math0 formalization of an OEIS conjecture and not part of formal-conjectures. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A211343.lean` in the directory `lean` after `lake build Math0.Statements.OEIS.A211343`; on our machine the former took well under a minute, and the latter, which includes the kernel data checks, about 5.5 minutes of wall time under heavy load (less than a minute of CPU time for the data checks).

References

[1] T. M. Apostol, *Sums of consecutive positive integers*, Math. Gazette **87** (2003), no. 508, 98–101.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171 [cs.AI], 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[5] J. J. Sylvester, *A constructive theory of partitions, arranged in three acts, an interact and an exodion*, Amer. J. Math. **5** (1882), 251–330.

[6] The OEIS Foundation Inc., OEIS entry A001227, `https://oeis.org/A001227`.

[7] The OEIS Foundation Inc., OEIS entry A109814 (D. W. Wilson), `https://oeis.org/A109814`.

[8] The OEIS Foundation Inc., OEIS entry A118235 (R. Zumkeller, 2006), `https://oeis.org/A118235`.

[9] The OEIS Foundation Inc., OEIS entry A196020 (O. E. Pol, 2013), `https://oeis.org/A196020`.

[10] The OEIS Foundation Inc., OEIS entry A211343 (O. E. Pol, 2013), `https://oeis.org/A211343`, revision #54.

[11] The OEIS Foundation Inc., OEIS entry A237048 (O. E. Pol, 2014), `https://oeis.org/A237048`.

[12] The OEIS Foundation Inc., OEIS entry A261699 (O. E. Pol, 2015), `https://oeis.org/A261699`.

[13] The OEIS Foundation Inc., OEIS entry A285898 (O. E. Pol and N. J. A. Sloane, 2017), `https://oeis.org/A285898`.

[14] The OEIS Foundation Inc., OEIS entry A286013 (O. E. Pol, 2017), `https://oeis.org/A286013`.

[15] The OEIS Foundation Inc., OEIS entry A379632 (O. E. Pol, 2024), `https://oeis.org/A379632`.